1. (a) True: $\because S_1, S_2$ are convex sets.

For $x, y \in (S_1 \cap S_2)$

$\because x, y \in S_1, \quad \therefore \forall \lambda \in [0,1], \lambda x + (1-\lambda) y \in S_1$

$\because x, y \in S_2 \quad \therefore \forall \lambda \in [0,1], \lambda x + (1-\lambda) y \in S_2$

$\therefore x, y \in S_1, \quad \therefore \forall \lambda \in [0,1], \lambda x + (1-\lambda) y \in (S_1 \cap S_2)$

$\therefore$ The intersection of two nonempty convex sets is again a convex set.

(b) False. A counter example:

$$A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 1 \\ 1 \\ -4 \end{bmatrix}$$

$$A_2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 1 & -1 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix}$$

$\therefore$ The graph →



$\therefore$ The union of P_1, P_2 is not a polyhedron

(c) False. A counter example.

$$c = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad A = [1, -1], \quad b = 0$$

$\therefore$ The optimal set is $\left\{ \begin{bmatrix} t \\ t \end{bmatrix} \mid t \geq 0 \right\}$, which is unbounded.

(d) False. A counter example:

$$A = [1\ 1], \quad b = 1, \quad c = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$\therefore$ minimize $x_1 + x_2$

s.t. $x_1 + x_2 = 1, \quad x_1, x_2 \geq 0$

$\therefore$ The objective function $x_1 + x_2 = 1$ is always constant over the feasible region.

$\therefore$ Every feasible point is optimal, which is not unique

$\therefore$ d) is False.

(e) True. On the graph, if we have more than one optimal solution, we can get an optimal line by connecting the two points. So that there are infinitely many optimal solutions.

$\vec{x} = r x_1 + (1-r) x_2$ is optimal for any $r \in [0,1]$

2. The constraints: $(x_1 - 1)^2 \leq 4 \Rightarrow |x_1 - 1| \leq 2$

$\max\{|x_1 + 1| - 2x_2, x_2\} \leq 2 \Rightarrow |x_1 + 1| - 2x_2 \leq 2, \quad x_2 \leq 2$

$x_1 - x_2 - x_3 \geq 0$

$\therefore \Rightarrow$ minimize $z - x_0^+ + x_0^-$

subject to $z - x_1^+ + x_1^- - 2x_2^+ + x_2^- - x_3^+ + x_3^- - s_1 = 0$

$z + x_1^+ - x_1^- + x_2^+ - x_2^- + x_3^+ - x_3^- - s_2 = 0$

$x_1^+ - x_1^- + s_3 - 2 = 0$

$x_2^+ - x_2^- - 1 - s_4 + 2^- = 0$

$x_3^+ - x_3^- - 2 + s_5 = 0$

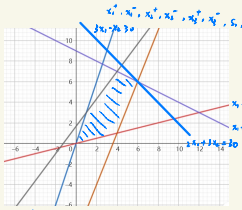
$x_1^+ - x_1^- + 1 - 2(x_2^+ + 2x_2^-) + s_6 = 0$

$-x_3^+ + x_3^- - 1 - 2x_2^+ + 2x_2^- - 2 + s_7 = 0$

$x_1^+ - x_1^- - x_2^+ + x_2^- - x_3^+ + x_3^- - s_8 = 0$

$x_1^+, x_1^-, x_2^+, x_2^-, x_3^+, x_3^-, s_1, s_2, s_3, s_4, s_5, s_6, s_7, s_8 \geq 0$

3.



Vertices: $(0,0)$, $(4,1)$, $(6,6)$, $(4,7)$, $(1,5)$

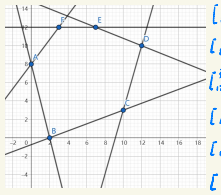
Active constraints:

$x_1 + 2x_2 \leq 18$

$3x_1 - 2x_2 \leq 18$.

Optimal solution: $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 6 \\ 6 \end{bmatrix}$, Optimal value: 30

4.



$\begin{bmatrix} 0 \\ 8 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} \rightarrow 4x + y \geq 8$

$\begin{bmatrix} 0 \\ 8 \end{bmatrix} \begin{bmatrix} 3 \\ 12 \end{bmatrix} \rightarrow 4x - 3y + 24 \geq 0$

$\begin{bmatrix} 3 \\ 12 \end{bmatrix} \begin{bmatrix} 7 \\ 12 \end{bmatrix} \rightarrow -y \geq -12$

$\begin{bmatrix} 7 \\ 11 \end{bmatrix} \begin{bmatrix} 12 \\ 10 \end{bmatrix} \rightarrow 2x + 5y - 74 \leq 0$

$\begin{bmatrix} 7 \\ 12 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \end{bmatrix} \rightarrow 7x - 3y - 64 \leq 0$

$\begin{bmatrix} 2 \\ 0 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \end{bmatrix} \rightarrow 3x - 8y - 6 \leq 0$

$\Rightarrow$ From the vertices of the graph, we can get the constraint edges of the linear problem. Then we can calculate the edges.

Assuming that s denotes the length of the square.

(x, y): The coordinate of the bottom-left point of the square.

$\therefore$ According to the graph:

maximize s

s.t. $4x + y \geq 8$

$4x - 3(y + s) + 24 = 4x - 3y - 3s + 24 \geq 0$

$y + s \leq 12$

$2(x + s) + 5(y + s) - 74 = 2x + 5y + 7s - 74 \leq 0$

$7(x + s) - 3y - 64 = 7x - 3y + 7s - 64 \leq 0$

$3x - 8y + 3s - 6 \leq 0$,

$x, y, s \geq 0$

$\Rightarrow$ By taking the bottom-left point of the starting point

Combine the coordinate of the four points and the constraint lines,

We can get the final constraints.

The four vertices of square should be within constrains and we want to maximize the length of the edge of square.

```
import cvxpy as cp
import numpy as np
import matplotlib.pyplot as plt


x = cp.Variable()
y = cp.Variable()
s = cp.Variable()

constraints = [
    4*x - 3*y - 3*s + 24 >= 0,
    y + s <= 12,
    2*x + 5*y + 7*s <= 74,
    7*x - 2*y + 7*s <= 64,
    3*x - 8*y + 3*s <= 6,
    4*x + y >= 8,
     x >= 0, y >= 0, s >= 0
]

problem = cp.Problem(cp.Maximize(s), constraints)
problem.solve()


print(f"x = {x.value:.5f}, y = {y.value:.5f}, s = {s.value:.5f}, square = {s.value**2:.5f}")

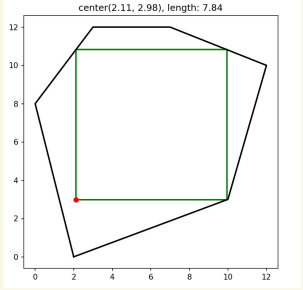
vertices = np.array([[2, 0], [10, 3], [12, 10], [7, 12], [3, 12], [0, 8], [2, 0]])


plt.figure(figsize=(6, 6))
plt.plot(vertices[:, 0], vertices[:, 1], 'k-', linewidth=2, label='edge')

x_opt, y_opt, s_opt = x.value, y.value, s.value
square = plt.Rectangle((x_opt , y_opt ), s_opt, s_opt,
                       edgecolor='green', facecolor='none', linewidth=2, label='optimal square')
plt.gca().add_patch(square)


plt.scatter(x_opt, y_opt, color='red', zorder=5, label='中心点')
plt.title(f"center({x_opt:.2f}, {y_opt:.2f}), length: { s_opt:.2f}")
plt.show()
```

```
PS C:\Users\lenovo> & C:/Users/lenovo/AppData/Local/Programs/Python/Python313/python.exe c:/Users/lenovo/Desktop/大二下/MAT3007/mat/test.py
x = 2.11446, y = 2.98193, s = 7.83735, square = 61.42405
```



5. The original problem :

$$\min_{x \in \mathbb{R}^2} \sum_{i=1}^{n} |a_i^T x - b_i| \quad \text{s.t. } \|x\|_1 \le \tau$$

Steps: We introduce an auxiliary variable to replace the absolute value.

we introduce $t_i \Rightarrow t_i \ge a_i^T x - b_i$ and $t_i \ge b_i - a_i^T x$, $\forall i$

$\therefore$ The objective function becomes $\sum_{i=1}^{n} t_i$

$\because \|x\|_1 = \sum_{i=1}^{n} |x_i|$ $\therefore$ we introduce z_i as another auxiliary variable.

$z_i \ge x_i$ and $z_i \ge -x_i$, $\forall i$

$\therefore$ The Linear Problem :

$$\begin{aligned}
&\text{minimize } \sum_{i=1}^{n} t_i \\
&x_i, t_i, z_i \\
&\text{subject to } t_i \ge a_i^T x - b_i \quad \forall i \\
&\qquad t_i \ge b_i - a_i^T x \quad \forall i \\
&\qquad z_i \ge x_i \quad \forall i \\
&\qquad z_i \ge -x_i \quad \forall i \\
&\qquad t_i , z_i \ge 0 \ , \forall i \\
&\qquad \sum_{i=1}^{n} z_i \le \tau
\end{aligned}$$

(b) $a_i = \begin{bmatrix} u_i \\ 1 \end{bmatrix}$, $b_i = v_i$

$\therefore$ minimize $\sum_{i=1}^{m} |a_i^T x - b_i|$ s.t. $\|x\|_1 \le \tau$
$x \in \mathbb{R}^2$

$\Rightarrow$ minimize $\sum_{i=1}^{m} |x_1 u_i + x_2 - v_i|$ s.t. $|x_1| + |x_2| \le \tau$
$x \in \mathbb{R}^2$

```python
import cvxpy as cp
import numpy as np
import matplotlib.pyplot as plt


D = np.array([[0, 0], [2, 0], [10, 3], [12, 10], [7, 12], [3, 12]])
u = D[:, 0]
v = D[:, 1]
m = len(D)

A = np.column_stack([u, np.ones(m)])  # a_i = [u_i, 1]
b = v                                 # b_i = v_i


tau_values = [0.1, 1, 10]
solutions = []

for tau in tau_values:

    x = cp.Variable(2)
    t = cp.Variable(m)


    constraints = [
        cp.norm(x, 1) <= tau,
        t >= A @ x - b,
        t >= -(A @ x - b),
        t >= 0
    ]


    obj = cp.sum(t)
    prob = cp.Problem(cp.Minimize(obj), constraints)
    prob.solve()

    solutions.append(x.value)

plt.figure(figsize=(12, 6))
plt.scatter(u, v, color='black', label='data point')
```

```python
plt.figure(figsize=(12, 6))
plt.scatter(u, v, color='black', label='data point')


u_plot = np.linspace(0, 12, 100)
for tau, x_opt in zip(tau_values, solutions):
    x1, x2 = x_opt
    v_fit = x1 * u_plot + x2
    plt.plot(u_plot, v_fit, label=f'τ={tau}, x1={x1:.2f}, x2={x2:.2f}')

plt.xlabel('u')
plt.ylabel('v')
plt.title(' τ ')
plt.legend()
plt.grid(True)
plt.show()
```

